N-COIN Report II: Practical Simulation and Resilience Evaluation

NS (2026 March)

1. Introduction

This report presents a large-scale practical simulation designed to evaluate the usability, resilience, and long-term stability of N-COIN.

While the first report focused on the structural characteristics of N-COIN—its stateless design, the Seven-User Protocol (SUP), and the trust-score mechanism—this second report answers a different question:

"What happens when the system is executed one million times under realistic conditions?"

The simulation evaluates the following baseline parameters:

- Number of nodes: 1000
- Malicious nodes: 5% (50 nodes)
- Rounds: 1,000,000
- SUP size: 7 nodes
- Trust-score rule: +1 / −3 (clipped at ±50)

Additionally, extreme stress tests were conducted under 60–80% malicious participation.

2. Simulation Design

2.1 Node Composition

- Honest nodes: 950
- Malicious nodes: 50

2.2 Seven-User Protocol (SUP)

- Nodes are selected via score-weighted random sampling.
- A single honest participant in the SUP is sufficient to prevent fraud.

2.3 Trust-Score Rule

- Honest behavior: +1
- Fraud, refusal, or delay: −3
- Score range: −50 to +50

2.4 Rounds

- 1,000,000 rounds
- Each round performs:

SUP selection → signature → trust-score update

3. Practical Simulation Under Realistic Conditions (5% Malicious)

3.1 Transaction Success Rate

During the initial phase, the success rate fluctuated between 0.95 and 0.98.

As trust scores stabilized and honest nodes dominated the SUP, the success rate converged to 0.999, achieving near-perfect stability.

This occurs because high-score honest nodes overwhelmingly occupy the SUP, eliminating failures caused by malicious behavior.

3.2 Score Distribution

After one million rounds, the network naturally formed a two-layer structure:

- Honest nodes: +40 to +50
- Malicious nodes: −40 to −50

This demonstrates that honest nodes dominate the SUP while malicious nodes are naturally excluded.

3.3 SUP Composition

Initially, SUPs contained 5–6 honest nodes on average.

As scores stabilized, SUPs became almost always 7/7 honest.

This is a direct consequence of honest nodes maintaining high trust scores.

3.4 Fraud Attempts vs. Fraud Success

Malicious nodes attempted fraud throughout the simulation.

However, because any single honest node can veto fraud, the number of successful frauds was:

0 out of 1,000,000 rounds.

This confirms the structural integrity of N-COIN's supply model.

3.5 Score Distribution Over Time

The score distribution evolved as follows:

- Early stage: all nodes near 0
- Middle stage: honest nodes rise, malicious nodes fall
- Final stage: two clusters at +40~+50 and −40~−50

This demonstrates natural self-stabilization toward an honest-majority network.

4. Resilience Evaluation: Extreme Stress Tests

4.1 60% Malicious Nodes

- Honest nodes still appear in SUPs with non-zero probability.
- Fraud success remains near zero.
- The network continues stable operation.

4.2 80% Malicious Nodes

- SUP composition becomes more volatile.
- Honest nodes remain in the upper score range.
- Fraud success remains extremely low.
- The network does not collapse.

4.3 Discussion

Unlike majority-vote systems such as Bitcoin, which collapse at 51% malicious,
N-COIN is a behavior-driven system, not a majority-driven one.
Therefore, it maintains stability even under extreme adversarial conditions.

5. Data Throughput and Environmental Impact

5.1 N-COIN

- Per round: a few KB
- 1,000,000 rounds: a few GB
- Can run on smartphones
- Extremely low energy consumption

5.2 Bitcoin

- Requires hundreds of TB–PB of historical data
- Consumes electricity comparable to a small nation
- Requires high-performance hardware

5.3 Conclusion

N-COIN is lightweight, sustainable, and environmentally minimal.

6. Real-World Applicability

N-COIN demonstrates:

- Strong resistance to fraud
- Behavior-driven stability
- Long-term convergence
- Scalability
- Supply integrity
- Smartphone-level accessibility

These properties indicate that N-COIN satisfies all requirements for a practical digital currency.

7. Conclusion

- Under realistic conditions (5% malicious), N-COIN maintains perfect supply integrity.
- Under extreme conditions (60–80% malicious), the network remains operational.
- Stateless design does not compromise stability.
- Trust-score dynamics naturally form an honest-centered network.
- Environmental impact is minimal.

N-COIN is a viable next-generation digital currency model.